

## Mini-Project

### Developing a Context-Aware Chatbot for Academic Support

#### Objective:

To design and implement a chatbot that provides academic support to university students by answering questions related to course content, providing information on academic deadlines, and offering guidance on university policies and resources.

#### Expected Outcomes:

- A fully functional chatbot capable of understanding complex queries in natural language
- Implement NLP techniques like entity extraction, response generation and intent recognition
- Evaluation of the chatbot performance through metrics such as accuracy, user satisfaction, and response time

#### Tasks:

1. **Literature Review:** Comprehensive overview of existing NLP models and techniques used for developing chatbots.
2. **Data Collection and Preprocessing:** Gather and preprocess data needed for training the chatbot. This may include student inquiries, FAQs, information about the study program, study material with respect to modules, examination office documents, and other relevant text data.
3. **Model Selection and Training:** Choose appropriate NLP models and train them on a prepared dataset. Adapt the models to improve context understanding and retention.
4. **Testing:** Test the chatbot with real users to get feedback and identify ideas of improvement
5. **Performance Evaluation:** Evaluate the chatbot using quantitative and qualitative methods to measure its effectiveness and accuracy.
6. **Documentation:** Document the development process and solutions implemented.